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System and method for attaching a trailer to a tow vehicle

Abstract

A trailer hitch system for attaching a trailer to a tow vehicle, is disclosed having a shank attachable a trailer, the shank having a nose at a terminal end of the shank and a mid-shank extending in a substantially horizontal direction immediately adjacent to the nose, the nose and mid-shank being configured to be received in a receiver attachable to said tow vehicle. The shank also includes a vertical adjustment extending in a substantially vertical direction, a depth stop connected to the shank and configured to abut the receiver to limit a depth that the shank extends within the receiver. The mid-shank includes step reduction which decreases a cross-sectional area of the mid-shank immediately adjacent to the nose, and the nose is configured to reduce in cross-sectional area toward the terminal end of the shank. The shank also includes an alignment indicator connected to the shank, such that the alignment indicator to provide a visual indicator of the orientation of the shank.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/436,722, filed Jun. 10, 2019, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/797,123, filed Jan. 25, 2019, and U.S. Provisional Patent Application Ser. No. 62/682,754, filed Jun. 8, 2018, which are hereby incorporated by reference herein in their entirety, including but not limited to those portions that specifically appear hereinafter, the incorporation by reference being made with the following exception: in the event that any portion of the above-referenced applications are inconsistent with this application, this application supercedes the above-referenced applications.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not Applicable.

BACKGROUND

1. The Field of the Present Disclosure

(2) The present disclosure relates generally to trailer hitch systems, and more particularly, but not necessarily entirely, to trailer hitch systems that facilitate connecting trailers to tow vehicles.

2. Description of Related Art

(3) Various different types of trailer hitches are known in the art for attaching trailers to towing vehicles. One of the most common types of towing systems utilizes a ball hitch. Typical ball hitches have a generally spherical shaped ball with a stem or threaded rod extending from a base of the ball. The threaded rod may be configured to engage a hitch head mounted on the towing vehicle using a threaded receiver or ball hitch fitting. A trailer coupling member, such as a ball clamp, positioned on a front of the trailer tongue, may engage the ball hitch in a loose friction fit, and may be secured to the ball in preparation for towing. Some hitches may also have load equalizing systems to distribute loads and allow the towing vehicle and trailer to remain level. Other mechanisms may also be used in association with a trailer hitch to reduce side sway of the trailer.

(4) A common challenge with the known trailer hitch systems is attaching the trailer to the tow vehicle. Difficulty may arise in properly aligning the trailer with the tow vehicle as the tow vehicle is backed to engage with the trailer. Often during attempts to attach the trailer to the tow vehicle,

the sharp rigid end of the trailer tongue may strike the back of the tow vehicle causing damage to the tow vehicle.

(5) There have been many attempts to improve the method of connecting trailers to tow vehicles, yet the most common method remains the conventional ball connection, even though such method has several drawbacks. One of the drawbacks of the conventional ball mount system and method may include the necessity to precisely position the hitch ball that is connected to the vehicle under the trailer tongue ball clamp. On weight distribution hitches, the difficulty of attaching the trailer may be increased by the common necessity of installing and removing the heavy ball mount assembly, spring bars and anti-sway bars if so equipped, every time the trailer is connected or disconnected.

(6) Other trailer hitch systems may include plug-in type hitches in which receivers are joined to the tow vehicle, and the hitch ball is joined to the trailer. A bar on the trailer tongue may be plugged into the receiver to attach the trailer to the tow vehicle. A drawback of the known plug-in hitches may include the requirement of a unique receiver hitch unit to be installed on the tow vehicle. Since vehicles used for towing are commonly equipped by the manufacturer with a receiver hitch already installed, removing and replacing the existing receiver hitch may be a significant impediment to the success of such towing systems. Other trailer hitch systems require unique mechanisms to be installed on the trailer. Yet other trailer hitch systems allow the plug-in receiver to be installed on the exterior of the manufacturer installed receiver, but such systems may add considerable cost, weight, and complexity to the trailer hitch.

(7) Some of the known plug-in type trailer hitch systems are prone to binding during engagement or disengagement of the trailer with the tow vehicle. Such binding may be caused by different heights of the receiver with respect to the connecting bar on the trailer tongue, misalignment, improperly shaped engaging elements, and/or the lack of adequate bearing surfaces or rolling elements to reduce friction between the receiver and the connecting bar on the trailer tongue. Because of the considerable weight of both the tow vehicle and the trailer, even small misalignments can generate considerable frictional forces.

(8) Various different mechanisms are known in the art for assisting in backing or aligning the tow vehicle with the trailer to facilitate attaching the trailer to the tow vehicle. Such mechanisms may be visual, optical, mechanical or electronic devices to assist in achieving the proper height, angle and position of the tow vehicle with respect to the trailer. Many of the devices are complex or bulky such that they may result in more of a hindrance than an aid.

(9) Despite the advantages of known trailer hitch systems, improvements are still being sought. The prior art is characterized by several disadvantages that are addressed by the present disclosure. The present disclosure minimizes, and in some aspects eliminates, the above-mentioned failures, and other problems, by utilizing the methods and structural features described herein.

(10) The features and advantages of the present disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by the practice of the present disclosure without undue experimentation. The features and advantages of the present disclosure may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base, or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each claim of this application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and advantages of the disclosure will become apparent from a consideration of the subsequent detailed description presented in connection with the accompanying drawings in which:

- (2) FIG. 1 is a perspective view of an embodiment of the disclosed invention;
- (3) FIG. 2 is a perspective view of another embodiment of the disclosed invention;
- (4) FIG. 3 is a side view of the embodiment of FIG. 2;
- (5) FIG. 4 is a perspective view of another embodiment of the disclosed invention;
- (6) FIG. 5 is a side view of the embodiment of FIG. 4;
- (7) FIG. 6 is a perspective view of another embodiment of the disclosed invention;
- (8) FIG. 7 is a side view of the embodiment of FIG. 6;
- (9) FIG. 8 is a perspective view of another embodiment of the disclosed invention;
- (10) FIG. 9 is a side view of the embodiment of FIG. 8;
- (11) FIG. 10 is a perspective view of another embodiment of the disclosed invention;
- (12) FIG. 11 is a side view of the embodiment of FIG. 10;
- (13) FIG. 12 is a perspective view of another embodiment of the disclosed invention;
- (14) FIG. 13 is a side view of the embodiment of FIG. 12;
- (15) FIG. 14 is a perspective view of another embodiment of the disclosed invention;
- (16) FIG. 15 is a side view of the embodiment of FIG. 14;
- (17) FIG. 16 is a perspective view of another embodiment of the disclosed invention;
- (18) FIG. 17 is a side view of the embodiment of FIG. 16;
- (19) FIG. 18 is a perspective view of another embodiment of the disclosed invention;
- (20) FIG. 19 is a partial cross-sectional view of the embodiment of FIG. 18;
- (21) FIG. 20 is a perspective view of another embodiment of the disclosed invention;
- (22) FIG. 21 is a partial cross-sectional view of the embodiment of FIG. 20;
- (23) FIG. 22 is a cross-sectional view of another embodiment of the disclosed invention;
- (24) FIG. 23 is a perspective view of the embodiment of FIG. 22;
- (25) FIG. 24 is a partial cut-away view of the embodiment of FIG. 22;
- (26) FIG. 25 is a perspective view of another embodiment of the disclosed invention;
- (27) FIG. 26 is a cross-sectional view of the embodiment of FIG. 25;
- (28) FIG. 27 is a perspective view of another embodiment of the disclosed invention;
- (29) FIG. 28 is a side view of the embodiment of FIG. 27;
- (30) FIG. 29 is a perspective view of another embodiment of the disclosed invention;
- (31) FIG. 30 is a cross-sectional view of the embodiment of FIG. 29;
- (32) FIG. 31 is a side view of another embodiment of the disclosed invention;
- (33) FIG. 32 is a perspective view of the embodiment of FIG. 31;
- (34) FIG. 33 is a perspective view of another embodiment of the disclosed invention;
- (35) FIG. 34 is a side view of the embodiment of FIG. 33;
- (36) FIG. 35 is a perspective view of another embodiment of the disclosed invention;
- (37) FIG. 36 is a side view of the embodiment of FIG. 35;
- (38) FIG. 37 is a perspective view of another embodiment of the disclosed invention;
- (39) FIG. 38 is a side view of the embodiment of FIG. 37;
- (40) FIG. 39 is a perspective view of another embodiment of the disclosed invention;
- (41) FIG. 40 is a front view of the embodiment of FIG. 39;
- (42) FIG. 41 is a perspective view of another embodiment of the disclosed invention;
- (43) FIG. 42 is a side view of another embodiment of the disclosed invention;
- (44) FIG. 43 is a perspective view of the embodiment of FIG. 42;
- (45) FIG. 44 is a perspective view of another embodiment of the disclosed invention;
- (46) FIG. 45 is a side view of another embodiment of the disclosed invention;
- (47) FIG. 46 is a perspective view of another embodiment of the disclosed invention;

- (48) FIG. 47 is a side view of the embodiment of FIG. 46;
- (49) FIG. 48 is a front view of the embodiment of FIG. 46;
- (50) FIG. 49 is a perspective view of another embodiment of the disclosed invention;
- (51) FIG. 50 is a side view of the embodiment of FIG. 49;
- (52) FIG. 51 is a perspective view of another embodiment of the disclosed invention;
- (53) FIG. 52 is a top view of the embodiment of FIG. 51;
- (54) FIG. 53 is a perspective view of another embodiment of the disclosed invention;
- (55) FIG. 54 is a top view of the embodiment of FIG. 51; and
- (56) FIG. 55 is a perspective view of another embodiment of the disclosed invention.

DETAILED DESCRIPTION

(57) For the purposes of promoting an understanding of the principles in accordance with the disclosure, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended. Any alterations and further modifications of the inventive features illustrated herein, and any additional applications of the principles of the disclosure as illustrated herein, which would normally occur to one skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the disclosure claimed.

(58) Before the present apparatus and methods for providing a trailer hitch system are disclosed and described, it is to be understood that this disclosure is not limited to the particular configurations, process steps, and materials disclosed herein as such configurations, process steps, and materials may vary somewhat. It is also to be understood that the terminology employed herein is used for the purpose of describing particular embodiments only and is not intended to be limiting since the scope of the present disclosure will be limited only by the appended claims and equivalents thereof.

(59) The publications and other reference materials referred to herein to describe the background of the disclosure, and to provide additional detail regarding its practice, are hereby incorporated by reference herein in their entireties, with the following exception: In the event that any portion of said reference materials is inconsistent with this application, this application supercedes said reference materials. The reference materials discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as a suggestion or admission that the inventors are not entitled to antedate such disclosure by virtue of prior disclosure, or to distinguish the present disclosure from the subject matter disclosed in the reference materials.

(60) It must be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Moreover, as used herein, the terms “comprising,” “including,” “containing,” “characterized by,” and grammatical equivalents thereof are inclusive or open-ended terms that do not exclude additional, unrecited elements or method steps.

(61) As used herein, the term “elongate” shall be construed broadly to include structures that have one dimension, such as a length, that is greater than another dimension, such as a width or diameter.

(62) As used herein, the term “proximal” shall refer broadly to the concept of a nearest portion. For example, the vertical adjustment is the proximal-most portion of the shank, because it is the nearest portion when said shank is installed.

(63) As used herein, the term “distal” shall generally refer to the opposite of proximal, and thus to the concept of a further portion, or a furthest portion, depending upon the context.

(64) As used herein, the general term “shank” shall generally refer to a shank used in a rise or drop position, as known in the art.

(65) The following inventive embodiments are related to a trailer hitch system for attaching a

trailer to a tow vehicle, where the trailer hitch system includes at least a shank attachable to a trailer and a tow vehicle.

(66) Referring now to FIG. 1, a first shank embodiment includes a shank **10**, having a nose **12** a mid-shank **14**, a support **16** and a vertical adjustment **18**. The mid-shank **14** can extend from the vertical adjustment **18**, or upright portion, to the nose **12**. The nose **12** may be defined as an end portion, extending to a terminating free end of the shank **10**, having a cross-sectional area less than the cross-sectional area of the mid-shank **14**. The nose **12**, as shown in FIG. 1, includes a tapered angular shape, which reduces in cross-sectional area until reaching a terminating free end of the nose **12**. By reducing the cross-sectional shape of the nose **12**, the nose **12** can improve the speed and time to install the shank **10** into a receiver, attachable to a tow vehicle, due to the tight tolerances between the exterior of the nose **12** and the interior of the receiver.

(67) While the tapered angular shape, shown in FIG. 1, includes a substantially square cross-section, various nose **12** geometries can be used with reduced cross-sectional areas, which can allow the shank **10** to initially enter into a receiver opening while allowing for some misalignment. A receiver, not shown, can be fixed, or otherwise attachable to a tow vehicle to enable the shank **10**, which can be connected to a trailer, to facilitate a removable connection between a tow vehicle and a trailer (also not shown).

(68) The support **16** can function to provide strength support to between the mid-shank **14** and the vertical adjustment, such that the mid-shank **14** and the vertical adjustment **18** may maintain a substantially 90 degree angles, or other, predetermined or desired angle. The vertical adjustment **18** can provide numerous adjustable positions, using holes **18a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(69) Referring now to FIGS. 2 and 3, another shank embodiment includes a shank **20**, having a nose **22** a mid-shank **24**, a support **26** and a vertical adjustment **28**, wherein the mid-shank **24**, support **26**, and vertical adjustment **28**, may be substantially the same as those shown and described in FIG. 1, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **28** can provide numerous adjustable positions, using holes **28a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(70) The nose **22** may be defined as an end portion, extending to a terminating free end of the shank **20**, having a cross-sectional area less than the cross-sectional area of the mid-shank **24**. The nose **22**, as shown in FIGS. 2 and 3, includes a rounded, substantially semicircular shape, which reduces in cross-sectional area until reaching a terminating free end of the nose **22**. By reducing the cross-sectional shape of the nose **22**, the nose **22** can improve the speed and time to install the shank **20** into a receiver, and thereby improve the initial entry of the shank **20** into the receiver by allowing the shank **20** to be out of alignment with the receiver, for example, up to a ½ inch out of alignment, or more, and still enter the receiver.

(71) Referring now to FIGS. 4 and 5, another shank embodiment includes a shank **30**, having a nose **32** a mid-shank **34**, a support **36** and a vertical adjustment **38**, wherein the mid-shank **34**, support **36**, and vertical adjustment **38**, may be substantially the same as those shown and described in FIG. 1, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **38** can provide numerous adjustable positions, using holes **38a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(72) The nose **32** may be defined as an end portion, extending to a terminating free end of the shank **30**, having a cross-sectional area less than the cross-sectional area of the mid-shank **34**. The nose **32**, as shown in FIGS. 3 and 4, includes a rounded perimeter, which reduces in cross-sectional area until reaching a terminating free end, which includes a substantially planar terminal end surface **32a**. By reducing the cross-sectional shape of the nose **32**, the nose **32** can improve the speed and time to install the shank **30** into a receiver, and thereby improve the initial entry of the

shank **30** into the receiver by allowing the shank **30** to be out of alignment with the receiver, for example, up to a inch out of alignment, or more, and still enter the receiver.

(73) Referring now to FIGS. **6** and **7**, another shank embodiment includes a shank **40**, having a nose **42** a mid-shank **44**, a support **46** and a vertical adjustment **48**, wherein the mid-shank **44**, support **46**, and vertical adjustment **48**, may be substantially the same as those shown and described in FIG. **1**, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **48** can provide numerous adjustable positions, using holes **48a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(74) The nose **42** may be defined as an end portion, extending to a terminating free end of the shank **40**, having a cross-sectional area less than the cross-sectional area of the mid-shank **44**. The nose **42**, as shown in FIGS. **3** and **4**, includes a rounded, substantially semi-circular or oblong shape, which reduces in cross-sectional area until reaching a terminating free end. By reducing the cross-sectional shape of the nose **42**, the nose **42** can improve the speed and time to install the shank **40** into a receiver, and thereby improve the initial entry of the shank **40** into the receiver by allowing the shank **40** to be out of alignment with the receiver, for example, up to a inch out of alignment, or more, and still enter the receiver.

(75) Referring now to FIGS. **8** and **9**, another shank embodiment includes a shank **50**, having a nose **52** a mid-shank **54**, a support **56** and a vertical adjustment **58**, wherein the mid-shank **54**, support **56**, and vertical adjustment **58**, may be substantially the same as those shown and described in FIG. **1**, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, The vertical adjustment **58** can provide numerous adjustable positions, using holes **58a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(76) The nose **52** may be defined as an end portion, extending to a terminating free end of the shank **50**, having a cross-sectional area less than the cross-sectional area of the mid-shank **54**. The nose **52**, as shown in FIGS. **8** and **9**, includes a rounded, cone-shaped perimeter, which reduces in cross-sectional area until reaching a terminating free end, which can come to a point or include a substantially planar terminal end surface **52a**. By reducing the cross-sectional shape of the nose **52**, the nose **52** can improve the speed and time to install the shank **50** into a receiver, and thereby improve the initial entry of the shank **50** into the receiver by allowing the shank **50** to be out of alignment with the receiver, for example, up to a inch out of alignment, or more, and still enter the receiver.

(77) Referring to FIGS. **10** and **11**, another shank embodiment includes a shank **100**, having a nose **122** a mid-shank **124**, a support **126** and a vertical adjustment **128**. The nose **122** is shown having a substantially straight, block-type configuration, however, various shapes or geometries can be used. The mid-shank **124** is shown having a decreased perimeter or cross-sectional area, via a step reduction **124a**, immediately adjacent to the nose **122**, which can allow the shank **100** to continue to more easily and completely enter into the receiver opening after the nose **122** initially enters into the receiver. The term “step reduction” as used herein and used through this specification, may include, but shall not be limited to, a 90 degree step or a right-angled corner. Various geometries of the mid-shank **124** can be used which can allow the shank **100** to easily and completely enter into a receiver opening.

(78) By reducing the cross-sectional area of the mid-shank **124**, the step reduction **124a** can improve the speed and time to install the shank **100** into a receiver, reducing, and possibly eliminating any binding that may take place between the shank **100** and the receiver, due to the tight tolerances between the exterior of the mid-shank **124** and the interior of the receiver. Specifically, the shank **100** may be attached or coupled to a trailer, prior to being received into the receiver. In this circumstance, a tow vehicle, having the receiver, may be backed into engagement with the shank **100** such that the receiver receives the shank **100** as the tow vehicle is backed up. If

the mid-shank **124** is not reduced in cross-sectional area, or waisted, as disclosed herein, the shank **100** can bind with the receiver, unless the shank **100** and the receiver are aligned within a tight tolerance.

(79) The support **126** can function to provide strength support to between the mid-shank **124** and the vertical adjustment, such that the mid-shank **124** and the vertical adjustment **128** may maintain a substantially 90 degree angles, or other, predetermined or desired angle. The vertical adjustment **128** can provide numerous adjustable positions, using holes **128a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(80) Referring now to FIGS. **12** and **13**, another shank embodiment includes a shank **200**, having a nose **222** a mid-shank **224**, a support **226** and a vertical adjustment **228**, wherein the nose **222**, mid-shank **224**, support **226**, and vertical adjustment **228**, may be substantially the same as those shown and described in FIG. **1** or **10**, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **228** can provide numerous adjustable positions, using holes **228a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(81) The mid-shank **224** is shown having a decreased perimeter or cross-sectional area, via a concave or curved reduction **224a**, immediately adjacent to the nose **222**, which can allow the shank **200** to continue to more easily and completely enter into the receiver opening after the nose **222** initially enters into the receiver. The curved reduction **224a** may, include four sides of the mid-shank **224** having the same, or substantially the same, curved reduction **224a**, or hour-glass shape. Various geometries of the mid-shank **224** can be used which can allow the shank **200** to easily and completely enter into a receiver opening.

(82) By reducing the cross-sectional area of the mid-shank **224**, the curved reduction **224a** can improve the speed and time to install the shank **200** into a receiver, reducing, and possibly eliminating any binding that may take place between the shank **200** and the receiver, due to the tight tolerances between the exterior of the mid-shank **224** and the interior of the receiver.

(83) Referring now to FIGS. **14** and **15**, another shank embodiment includes a shank **300**, having a nose **322** a mid-shank **324**, a support **326** and a vertical adjustment **328**, wherein the nose **322**, mid-shank **324**, support **326**, and vertical adjustment **328**, may be substantially the same as those shown and described in FIG. **1** or **10**, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **328** can provide numerous adjustable positions, using holes **328a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(84) The mid-shank **324** is shown having a decreased perimeter or cross-sectional area, via a reduction **324a**, which may be stepped or rounded until merging with a substantially flat or planar portion **324c**, immediately adjacent to the nose **322**. The substantially planar portion **324c** may be substantially parallel with the remaining portion of the mid-shank **324**. The mid-shank **324** may also include an increase **324b**, which may be stepped or rounded until merging with the substantially planar portion **324c**, proximal to the reduction **324a**. The combination of the reduction **324a** and the increase **324b** provides a section of decreased cross-sectional area, compared to the remaining portion of the mid-shank **324**, which can allow the shank **300** to continue to more easily and completely enter into the receiver opening after the nose **322** initially enters into the receiver. The step reduction **324a** and step increase Various geometries of the mid-shank **324** can be used which can allow the shank **300** to easily and completely enter into a receiver opening.

(85) By reducing the cross-sectional area of the mid-shank **324**, the reduction **324a** and increase **324b**, can improve the speed and time to install the shank **300** into a receiver, reducing, and possibly eliminating any binding that may take place between the shank **300** and the receiver, due to the tight tolerances between the exterior of the mid-shank **324** and the interior of the receiver.

(86) Referring now to FIGS. **16** and **17**, another shank embodiment includes a shank **400**, having a nose **422** a mid-shank **424**, a support **426** and a vertical adjustment **428**, wherein the nose **422**, mid-

shank **424**, support **426**, and vertical adjustment **428**, may be substantially the same as those shown and described in FIG. **1** or **10**, or of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. Additionally, the vertical adjustment **428** can provide numerous adjustable positions, using holes **428a**, which can receive a pin for attachment of a ball hitch or other hitch attachment.

(87) The mid-shank **424** is shown having a decreased perimeter or cross-sectional area, via a concave or curved reduction **424a**, immediately adjacent to the nose **422**, which can allow the shank **400** to continue to more easily and completely enter into the receiver opening after the nose **422** initially enters into the receiver. The curved reduction **424a** may include a double conical shape where the cross-section area reduces immediately adjacent to the nose **422**, until the center of the curved reduction **424a**, and then increase towards the remainder of the mid-shank **424**. Various geometries of the mid-shank **424** can be used which can allow the shank **400** to easily and completely enter into a receiver opening.

(88) By reducing the cross-sectional area of the mid-shank **424**, the curved reduction **424a** can improve the speed and time to install the shank **400** into a receiver, reducing, and possibly eliminating any binding that may take place between the shank **400** and the receiver, due to the tight tolerances between the exterior of the mid-shank **424** and the interior of the receiver.

(89) Referring to FIGS. **18** and **19**, another shank embodiment includes a shank **1000**, partially inserted into a receiver **1012**. The shank **1000** also includes a mid-shank **1014**, a depth stop **1016** and a vertical adjustment **1018**, wherein the mid-shank **1014** and vertical adjustment **1018**, and nose (not shown) may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The depth stop **1016** can contact the receiver **1012** when the desired shank **1000** depth has been reached. The depth stop **1016** can be adjustable to alter the desired depth of the shank **1000** within the receiver **1012**. This adjustability can be facilitated by a threaded member **1017a** which can be received and threadedly engaged with a receiving portion **1017b**, which can enable the threaded member **1017a** to extend in a proximal/distal, or lateral, direction towards or away from the receiver **1012**.

(90) The threaded member **1017a** may also include a flat or blunt end **1017c** which can be configured to abut a rim **1012a** of the receiver **1012**, thereby restricting or limiting the depth that the shank **1000** can enter into the receiver **1012**. Although only a threaded adjustable depth stop **1016** is shown in FIGS. **18** and **19**, alternative depth stops can be used which can limit the depth of the shank **1000** in the receiver **1012**, and such depth stops can be fixed in position or adjustable.

(91) Referring to FIGS. **20** and **21**, another shank embodiment includes a shank **1100**, partially inserted into a receiver **1112**. The shank **1100** also includes a mid-shank **1114**, a depth stop **1116** and a vertical adjustment **1118**, wherein the mid-shank **1114** and vertical adjustment **1118**, and nose (not shown) may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(92) The depth stop **1116** can contact the receiver **1112** when the desired shank **1100** depth has been reached. The depth stop **1116** may be formed as a collar having an increased diameter when compared to the mid-shank **1114** and can be adjustably secured or coupled to the mid-shank **1114** via bolts or fasteners **1117**, which enables the depth stop **1116** to alter or change the depth of the shank **1100** within the receiver **1112** or the position of the depth stop **1116** with respect to the mid-shank **1114**.

(93) The depth stop **1116** may have a substantially planer or blunt surface which can be configured to abut a rim **1112a** of the receiver **1112**, thereby restricting or limiting the depth that the shank **1100** can enter into the receiver **1112**.

(94) Referring to FIGS. **22-24**, another shank embodiment includes a shank **1200**, partially inserted into a receiver **1212**. The shank **1200** also includes a mid-shank **1214**, a depth stop **1216** and a vertical adjustment **1218**, wherein the mid-shank **1214** and vertical adjustment **1218**, and nose (not shown) may be of the form or shape, or combination of form or shape, of any or the shank

embodiments disclosed herein.

(95) The depth stop **1216** can contact the receiver **1212** when the desired shank **1200** depth has been reached. The depth stop **1216** be formed as a collar having an increased diameter when compared to the mid-shank **1214** and can fixed or secured to the mid-shank **1214**, at a desired depth of the shank **1200** within the receiver **1212**.

(96) The depth stop **1216** may have a substantially planer or blunt surface which can be configured to abut a rim **1212a** of the receiver **1212**, thereby restricting or limiting the depth that the shank **1200** can enter into the receiver **1212**.

(97) Referring to FIGS. **25** and **26**, another shank embodiment includes a shank **1300**, partially inserted into a receiver **1312**. The shank **1300** also includes a mid-shank **1314**, a depth stop **1316** and a vertical adjustment **1318**, wherein the mid-shank **1314** and vertical adjustment **1318**, and nose (not shown) may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(98) The depth stop **1316** can contact the receiver **1312** when the desired shank **1300** depth has been reached. The depth stop **1316** be formed as a plate essentially increasing the diameter of the mid-shank **1314**, and the plate positioning can be adjustably secured to the mid-shank **1314** via bolts or fasteners **1317**, to alter the desired depth of the shank **1300** within the receiver **1312**. The depth stop **1316** may have a substantially planer or blunt surface which can be configured to abut a rim **1312a** of the receiver **1312**, thereby restricting or limiting the depth that the shank **1300** can enter into the receiver **1312**.

(99) Referring to FIGS. **27** and **28**, another shank embodiment includes a shank **1400**, partially inserted into a receiver **1412**. The shank **1400** also includes a mid-shank **1414**, a depth stop **1416** and a vertical adjustment **1418**, wherein the mid-shank **1414** and vertical adjustment **1418**, and nose (not shown) may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(100) The depth stop **1416** can contact the receiver **1412** when the desired shank **1400** depth has been reached. The depth stop **1416** be formed as a pin which can be inserted into and laterally fixed to the mid-shank **1414**, to alter the desired depth of the shank **1400** within the receiver **1412**. The depth stop **1416** may be configured to abut a rim **1412a** of the receiver **1412**, thereby restricting or limiting the depth that the shank **1400** can enter into the receiver **1412**.

(101) Referring to FIGS. **29** and **30**, another shank embodiment includes a shank **1500**, partially inserted into a receiver **1512**. The shank **1500** also includes a nose **1513**, mid-shank **1514**, a depth stop **1516** and a vertical adjustment **1518**, wherein the nose **1513**, mid-shank **1514**, vertical adjustment **1518**, may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(102) The depth stop **1516** can be formed as a plug which can be inserted into the receiver **1512** and attached to the receiver at a desired depth. The depth stop **1516** may be attached to the receiver via a pin (not shown) inserted into hole **1516a** which extends through both the receiver **1512** and the depth stop **1516**, or by any other suitable attachment mechanism. The depth stop **1516**, when attached to the receiver **1512**, will limit or restrict the depth that the shank **1500** can enter into the receiver **1512**, by abutting the nose **1513** of the shank **1500**.

(103) Referring to FIGS. **31** and **32**, another shank embodiment includes a shank **2000**, having a nose **2012** a mid-shank **2014**, a support **2016** and a vertical adjustment **2018**, wherein the mid-shank **2014** and vertical adjustment **2018**, and nose **2012** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2000** also includes an alignment indicator **2019**, shown as a horizontal, or substantially horizontal, stripe which extends substantially linearly, or substantially horizontally, along the mid-shank **2014** from the vertical adjustment **2018** to the nose **2012**. The term “substantially horizontally,” as used herein and throughout this specification, may include horizontal or a position or direction that is within 15 degrees of horizontal, or 15 degrees from horizontal. The alignment indicator **2019** may be

reflective, or at least more reflective than the rest of the shank **2000**, including the nose **2012** and the mid-shank **2014**, and/or a different color, such as red, yellow, or orange, for example, than the rest of the shank **2000**, including the nose **2012** and mid-shank **2014**, to more easily distinguish it from the rest of the shank **2000**, and enable a user to visually identify the orientation of the shank **2000**, such that the alignment indicator can provide a visual indicator of the orientation of the shank **200**. The alignment indicator may also be recessed along the mid-shank **2014**, which can reduce the likelihood that the alignment indicator **2019** may be scraped off the mid-shank **2014** during use. The shank **2000**, for example, may be inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2000**.

(104) The alignment indicator **2019** may also be present on each side of the mid-shank **2014**, including top and bottom surfaces. Alternatively, the alignment indicator **2019** can be present on a single side of the mid-shank **2014** or any desired combination of sides, including only the top and bottom surfaces of the mid-shank **2014**. Additionally, the alignment indicator **2019** can be removably attached to the mid-shank **2014**, permanently fixed thereto, or integrated with the mid-shank **2014**. Therefore, the alignment indicator **2019** can help a driver of the vehicle to more easily see the alignment of the shank **2000**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2000** into the receiver. The alignment indicator **2019** can alternatively be a structural or textured feature of the shank **2000**, which can also run substantially parallel with the mid-shank **2014**, in the same way the horizontal stripe of the alignment indicator **2019** runs substantially parallel to the mid-shank **2014**.

(105) Referring to FIGS. **33** and **34**, another shank embodiment includes a shank **2100**, having a nose **2112** a mid-shank **2114**, a support **2116** and a vertical adjustment **2118**, wherein the mid-shank **2114** and vertical adjustment **2118**, and nose **2112** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2100** also includes an alignment indicator **2119**, shown as multiple horizontal channels creating raised horizontal stripes (as shown on the side surface of the mid-shank **2114**), or a single horizontal channel (as shown on the top surface of the mid-shank **2114**) which extends substantially linearly from the vertical adjustment **2118** to the nose **2112**. The alignment indicator **2119** may be a different color or more reflective than the rest of the shank **2100**, in a similar or identical manner as discussed in previously discussed embodiments, and may be integrally formed into the shank **2100**, to more easily distinguish it from the rest of the shank **2100**. The shank **2100**, for example, may be inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2100**.

(106) The alignment indicator **2119** may also be present on each side of the mid-shank **2114**, or alternatively, the alignment indicator **2119** can be present on a single side of the mid-shank **2114** or any desired combination of sides. Therefore, the alignment indicator **2119** can help a driver of the vehicle to more easily see the alignment of the shank **2100**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2100** into the receiver.

(107) Referring to FIGS. **35** and **36**, another shank embodiment includes a shank **2200**, having a nose **2212** a mid-shank **2214**, a support **2216** and a vertical adjustment **2218**, wherein the mid-shank **2214** and vertical adjustment **2218**, and nose **2212** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2200** also includes an alignment indicator **2219**, shown as light or laser pointer, emitting a linear beam of light **2219a** from the alignment indicator **2219** toward a receiver (not shown). The alignment indicator **2219** may be integrally formed, fixed or removably attached, to the shank **2200**, and enable a driver to align the receiver with the shank **2200**, by following the emitted light beam **2219a**. Alternatively, the alignment indicator **2219** may be integrally formed, fixed or removably attached, to the receiver, pointing the light beam **2219a** toward the shank **2200**. In another embodiment, the alignment indicator **2219** may, include a cord retraction device which can retain a

cord (the cord may instead constitute a flexible cable or any suitable flexible strand), which can be connected to the receiver, which can be retracted into the alignment indicator **2219** as the receiver approaches the shank **2200**. The cord can then enable a driver to align the receiver with the shank **2200**, by following the cord. The shank **2200**, for example, may then be inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2200**.

(108) Therefore, the alignment indicator **2219** can help a driver of the vehicle to more easily see the alignment of the shank **2200**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2200** into the receiver.

(109) Referring to FIGS. **37** and **38**, another shank embodiment includes a shank **2300**, having a nose **2312** a mid-shank **2314**, a support **2316** and a vertical adjustment **2318**, wherein the mid-shank **2314** and vertical adjustment **2318**, and nose **2312** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2300** also includes an alignment indicator **2319**, shown as light emitting device, emitting light from the alignment indicator **2319** illuminating the area surrounding, and including, the shank **2314** (particularly in low light conditions). The alignment indicator **2319** may be integrally formed, fixed, or removably attached, to the shank **2300** and enable a driver to better see and align the receiver with the shank **2300**. The shank **2300**, for example, may be inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2300**.

(110) Therefore, the alignment indicator **2319** can help a driver of the vehicle to more easily see the alignment of the shank **2300**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2300** into the receiver.

(111) Referring to FIGS. **39** and **40**, another shank embodiment includes a shank **2400**, having a nose **2412** a mid-shank **2414**, a support **2416** and a vertical adjustment **2418**, wherein the mid-shank **2414** and vertical adjustment **2418**, and nose **2412** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2400** also includes an alignment indicator **2419**, shown as an indent, carve-out, or recessed feature, centered on the terminal end or surface of the nose **2412**. The alignment indicator **2419** may also be formed as a different color than the rest of the shank **2400**, including the nose **2412** and mid-shank **2414**, such as red, yellow, or orange, for example, or may be more reflective than the rest of the shank **2400**, in a similar or identical manner as discussed in previously discussed embodiments. The alignment indicator **2419** may be integrally formed into the shank nose **2412** and enable a driver to align the receiver with the shank **2400**. As shown in FIGS. **39** and **40**, the alignment indicator **2419**, may be substantially square-shaped, or alternatively, the alignment indicator **2419** may be circular, triangular, star-shaped, or any other desired shape. Thus, the alignment indicator **2419** may aide a user to align the shank **2400**, for example, as the shank **2400** is inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2400**.

(112) Therefore, the alignment indicator **2419** can help a driver of the vehicle to more easily see the alignment of the shank **2400**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2400** into the receiver.

(113) Referring to FIGS. **40** and **41**, another shank embodiment includes a shank **2500**, having a nose **2512** a mid-shank **2514**, a support **2516** and a vertical adjustment **2518**, wherein the mid-shank **2514** and vertical adjustment **2518**, and nose **2512** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein. The shank **2500** also includes an alignment indicator **2519**, shown as an indent, carve-out, or recessed feature, centered on the terminal end or surface of the nose **2412**, and an additional alignment indicator located on an upper surface of the shank **2500**. The alignment indicator **2519** may also be formed as a different color than the rest of the shank **2500**, including the nose **2512** and mid-shank **2514**, such as red, yellow, or orange, for example, or may be more reflective than the rest of the shank **2500**, in a similar or identical manner as discussed in previously discussed embodiments. The alignment indicator **2519** may be integrally formed into the shank nose **2412** and enable a driver to

align the receiver with the shank **2400**. As shown in FIGS. **41** and **42**, the alignment indicator **2519**, may be substantially square-shaped, or alternatively, the alignment indicator **2519** may be circular, triangular, star-shaped, or any other desired shape. Thus, the alignment indicator **2519** may aide a user to align the shank **2500**, for example, as the shank **2500** is inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **2500**.

(114) Therefore, the alignment indicator **2419** can help a driver of the vehicle to more easily see the alignment of the shank **2400**, using rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **2400** into the receiver.

(115) Referring to FIG. **41**, another shank embodiment includes a shank **3000**, having a nose **3012**, a mid-shank **3014**, a support **3016**, a vertical adjustment **3018**, and an alignment indicator **3019**, wherein the mid-shank **3014**, vertical adjustment **3018**, nose **3012** and alignment indicator **3019**, may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(116) The shank **3000** is attached to a hitch **3020** which is mounted to weight distribution spring bars **3022**, to adjust the weight distribution of the trailer with respect to the to hitch **3020** after the shank **3000**, including the nose **3012** and mid-shank **3014**, is received into the receiver. The hitch **3020** is shown coupled to a coupler **3024** of a trailer (not shown). In use, the shank **3000** needs to be nearly horizontal (slight variances are acceptable) to allow the tow vehicle to back the receiver into engagement with the shank **3000**. As shown in FIG. **41**, the hitch **3020** is connected to the spring bars **3022** which are secured to a trailer frame **3026**, via brackets **3028**. The engagement between the spring bars **3022**, trailer frame **3026** and the shank **3000**, enable the shank **3000** to be held in a substantially horizontal position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(117) Referring to FIG. **42**, another shank embodiment includes the shank **3000**, hitch **3020** and trailer frame **3026** (and corresponding features) of FIG. **41**, however, this embodiment includes a spring bar stand **3030** which can be used to support the spring bars **3022** in a desired position. Thus, the engagement between the spring bars **3022**, spring bar stand **3030**, and the shank **3000**, enable the shank **3000** to be held in a substantially horizontal position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(118) Referring to FIG. **43**, another shank embodiment includes the shank **3000**, hitch **3020** and trailer frame **3026** (and corresponding features) of FIG. **41**, however, this embodiment includes a shank stand **3032** which can be used to support the shank **3000** in a desired position. Thus, the engagement between the shank **3000** and shank stand **3032**, enable the shank **3000** to be held in a substantially horizontal position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(119) Referring to FIG. **44**, another shank embodiment includes the shank **3000**, hitch **3020** and trailer frame **3026** (and corresponding features) of FIG. **41**, however, this embodiment includes a spring bar band **3034** which can be used to support the spring bars **3022** in a desired position. Thus, the engagement between the shank **3000**, the hitch **3020**, the spring bars **3022** and the spring bar band **3034**, enable the shank **3000** to be held in a substantially horizontal position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(120) Referring to FIG. **45**, another shank embodiment includes the shank **3000**, hitch **3020** and trailer frame **3026** (and corresponding features) of FIG. **41**, however, this embodiment includes a hitch band **3036** which can be used to support the hitch **3020** in a desired position. Thus, the engagement between the shank **3000**, the hitch **3020**, the hitch band **3036**, enable the shank **3000** to be held in a substantially horizontal position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(121) Referring to FIG. **46-48**, another shank embodiment includes the shank **3000**, hitch **3020** and

trailer frame **3026** (and corresponding features) of the embodiment of FIG. **41**, however, this embodiment includes an adjustment member or spring bar chain **3040** (a pair of adjustment members are used, although only one can be seen in FIGS. **46-48**, corresponding to the pair of spring bars **3022**), which can be used to support the spring bars **3022** in a desired position. The spring bar chains **3040** are flexible and adjustably coupled to the trailer frame **3026** via the corresponding brackets **3028**, the brackets **3028** having an adjustment feature **3028a** which can receive and adjust the position and length of the spring bar chains **3040** and removably couple the spring bar chains **3040** to the brackets **3028**.

(122) The spring bar chains **3040** may be removably coupled to the spring bars **3022** via a ring **3040a** which can receive and support the spring bars **3022**, prior to the shank **3000** being received into the receiver. Thus, the engagement between the shank **3000**, the hitch **3020**, the spring bars **3022**, and spring bar chains **3040**, enable the shank **3000** to be held in a substantially horizontal position, or easily adjust the horizontal portion (mid-shank) of the shank **3000** to a level position, even prior to engagement with the receiver, thus aiding in an increased speed and accuracy of engaging the shank **3000** with the receiver.

(123) Referring to FIGS. **49** and **50**, another shank embodiment includes a shank **4000**, having a nose **4012** a mid-shank **4014**, a support **4016** and a vertical adjustment **4018**. The nose **4012** is shown having a tapered angular shape, with chamfer edges, which reduces in cross-sectional area until reaching a terminating free end of the nose **4012**. The terminating end of the nose **4012** includes an alignment indicator **4019** which includes an indent, recess, or carve-out, centered on the terminal end surface of the nose **4012**. The alignment indicator **4019** may be integrally formed into the shank nose **4012**, or in any manner discussed above with respect to other embodiments, and enable a driver to align a receiver with the shank **4000**.

(124) The mid-shank **4014** is shown having a decreased perimeter or cross-sectional area, via a step reduction **4014a**, immediately adjacent to the nose **4012**, which can allow the shank **4000** to continue to more easily and completely enter into the receiver opening after the nose **4012** initially enters into the receiver. Various geometries of the mid-shank **4014** can be used which can allow the shank **4000** to easily and completely enter into a receiver opening. The step reduction **4014a** may be have a continuously sloped portion extending until reaching the greatest width of the mid-shank **4014**. The continuously sloped portion of the step reduction may be 0.125 inches in length L, or 0.1 to 0.5 inches, for example. The angle B of the sloped portion of the step reduction may be 2 to 10 degrees, or 4 degrees, for example. This continuously sloped portion can provide a shallower angle, while maintaining the same, or better, binding avoidance characteristics, as a circular scallop having max slope of about 15 degrees, for example. This continuously sloped portion of the step reduction **4014a** can thus avoid binding within a receiver that is about 10 degrees, or more, out of direct alignment with the shank **4000**, than a comparable circular scallop having a 15 degree maximum angle, for example.

(125) A terminal surface of the nose **4012a** may have a height or width of 0.5 to 1.5 inches, or 1 inch, for example. The nose **4012** may then extend at an angle A from the terminal surface **4012a**. The angle A may be less than 45 degrees, 25 to 40 degrees, or 30 degrees, for example. The angle A may cause binding of the shank **4000** as it is inserted into a receiver if the angle A is too steep, for example if the angle A is 45 degrees or more. The nose **4012** may increase in width from the terminal surface **4012a** to a its greatest width, which may be 1.5 to 3 inches, or two inches, for example.

(126) The mid-shank **4014** may also include a pin hole **4015** which may be positioned behind the step reduction **4014a** in a portion of the mid-shank having a constant width. The pin hole **4015** may have any desired diameter, for example, 1 inch, and may be configured to receive a retaining pin which can be inserted through the receiver, as a means of retaining the shank **4000** within the receiver. This configuration may provide a stronger pin connection, as the majority of the forces acting on a pin within the pin hole **4015** will be in shear, as opposed to a bending stress.

(127) The vertical adjustment **4018** can provide numerous adjustable positions, using holes **4018a**, which can receive a pin **4050** for attachment of a ball hitch **4052** or other hitch attachment, including, but not limited to, standard or non-height adjustable ball mounts.

(128) The shank **4000** also includes a depth stop **4017** which can contact a receiver when the desired shank **4000** depth has been reached. The depth stop **4017** can be adjustable to alter the desired depth of the shank **4000** within the receiver. This adjustability can be facilitated by a threaded member **4017a** which can be received and threadedly engaged with a receiving portion **4017b**, which can enable the threaded member **4017a** to extend in a proximal/distal, or lateral, direction towards or away from the receiver, along the mid-shank **4014**.

(129) Referring to FIGS. **51** and **52**, another shank embodiment includes a shank **5000**, having a nose **5012** a mid-shank **5014**, a support **5016** and a vertical adjustment **5018**. The nose **5012** is shown having a tapered angular shape, with chamfer edges, which reduces in cross-sectional area until reaching a terminating free end of the nose **5012**. The terminating end of the nose **5012** includes an alignment indicator **5019** which includes an indent, recess, or carve-out, centered on the terminal end surface of the nose **5012**. The alignment indicator **5019** may be integrally formed into the shank nose **5012**, or in any manner discussed above with respect to other embodiments, and enable a driver to align a receiver **6000** with the shank **5000**. A second alignment indicator **5020** may be connected or otherwise incorporated on a top surface of the mid-shank **5014**. The second alignment indicator **5020** may be linear or substantially linear, or may be substituted with any other alignment indicated disclosed herein.

(130) The mid-shank **5014** is shown having a decreased perimeter or cross-sectional area, via a step reduction **5014a**, immediately adjacent to the nose **5012**, which can allow the shank **5000** to continue to more easily and completely enter into the receiver opening **6002** after the nose **5012** initially enters into the receiver **6000**. Various geometries of the mid-shank **5014** can be used which can allow the shank **5000** to easily and completely enter into a receiver **6002** opening.

(131) The vertical adjustment **5018** can provide numerous adjustable positions, using holes **5018a**, which can receive a pin (not shown) for attachment of a ball hitch, for example, or other hitch attachment, including, but not limited to, standard or non-height adjustable ball mounts.

(132) The receiver **6000** may be attached to a tow vehicle, not shown, but may be visible to a driver of the tow vehicle via a back-up or rear facing camera (not shown), which may conventionally be attached to a rear bumper of the tow vehicle or at another rear-facing location on the tow vehicle. The receiver may include an alignment indicator **6004** on a top surface or portion of the receiver **6000** and positioned centered, or substantially centered, above the receiver opening **6002**. The alignment indicator **6004** on the receiver **6000** may be a different color (any desired color) than the rest of the receiver **6000** and may be formed as any desired shape. The alignment indicator **6004** may also be integral with the receiver **6000**, affixed thereto, using paint, tape or glue, for example. The receiver alignment indicator **6004** may also be aligned, collinear or substantially collinear, with the shank alignment indicator **5019** on the nose **5012** and the second alignment indicator **5020** on the mid-shank **5014**, when the shank **5000** is received within the opening **6002** of the receiver **6000**. This configuration and alignment of the alignment indicators **5019**, **5019** and **6004** enables the driver of the tow vehicle to more easily navigate and align the receiver opening **6002** into engagement with the shank **5000**, while operating the vehicle. Accordingly, the alignment indicator **6004** on the receiver **6004** may be visible to the tow vehicle driver via the back-up camera.

(133) In an alternative embodiment as shown in FIGS. **53** and **54**, for example, a tow vehicle bumper **7000**, may also include an alignment indicator **7002** on a top portion of the bumper **7000** in addition to, or instead of, having the alignment indicator **6004** on the receiver **6000**, particularly if the alignment indicator **6004** on the receiver **6002** is not visible to the tow vehicle operator via the back-up camera. The alignment indicator **7002** on the bumper **7000** may be a different color (any desired color) than the rest of the bumper **7000** and may be formed as any desired shape. The alignment indicator **7002** may also be integral with the bumper **7000**, affixed thereto, using paint,

tape or glue, for example. The alignment indicator **7002** on the bumper **7000** may also be aligned, collinear or substantially collinear, with the shank alignment indicator **5019** on the nose **5012** and the second alignment indicator **5020** on the mid-shank **5014**, when the shank **5000** is received within the opening **6002** of the receiver **6000**. This configuration and alignment of the alignment indicators **5019**, **5019**, **6004** and **7002** enables the driver of the tow vehicle to more easily navigate and align the receiver opening **6002** into engagement with the shank **5000**, while operating the vehicle. Accordingly, the alignment indicator **7002** on the bumper **7000** may be visible to the tow vehicle driver via the back-up camera.

(134) In alternative embodiments a colored alignment indicator may be affixed to a display of the back-up camera within the tow vehicle, thereby providing a reference point indicating when the shank is substantially aligned with the receiver. Additional or alternatively, an alignment indicator may be integrated with the edge of the display or an imbedded electronic marker may be programmed into the back-up camera system, and visible on the display within the vehicle, providing a reference point indicating when the shank is substantially aligned with the receiver.

(135) Referring to FIG. 55, another shank embodiment includes a shank **8000**, having a nose **8012** a mid-shank **8014**, a support **8016** and a vertical adjustment **8018**, wherein the mid-shank **8014** and vertical adjustment **8018**, and nose **8012** may be of the form or shape, or combination of form or shape, of any or the shank embodiments disclosed herein.

(136) The vertical adjustment **8018** can provide numerous adjustable positions, using holes **8018a**, which can receive a pin (not shown) for attachment of a ball hitch, for example, or other hitch attachment, including, but not limited to, standard or non-height adjustable ball mounts.

(137) The shank **8000** also includes an alignment indicator **8019**, shown as a plurality of dots or perforations, extending substantially linearly from the vertical adjustment **8018** to the nose **8012**. The alignment indicator **8019** may be a different color or more reflective than the rest of the shank **8000**, in a similar or identical manner as discussed in previously discussed embodiments, and may be integrally formed into the shank **8000**, to more easily distinguish it from the rest of the shank **8000**. The shank **8000**, for example, may be inserted into a receiver (not shown) of a vehicle, as the vehicle is reversed into engagement with the shank **8000**.

(138) Therefore, the alignment indicator **8019** can help a driver of the vehicle to more easily see the alignment of the shank **8000**, using a rear view mirror or a “back-up camera”, for example, and thereby, improve the speed and efficiency of inserting the shank **8000** into the receiver.

(139) In the foregoing Detailed Description, various features of the present disclosure are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed disclosure requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the following claims are hereby incorporated into this Detailed Description of the Disclosure by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

(140) It is to be understood that the above-described arrangements are only illustrative of the application of the principles of the present disclosure. Numerous modifications and alternative arrangements may be devised by those skilled in the art without departing from the spirit and scope of the present disclosure and the appended claims are intended to cover such modifications and arrangements. Thus, while the present disclosure has been shown in the drawings and described above with particularity and detail, it will be apparent to those of ordinary skill in the art that numerous modifications, including, but not limited to, variations in size, materials, shape, form, function and manner of operation, assembly and use may be made without departing from the principles and concepts set forth herein.

Claims

1. A trailer hitch system for attaching a trailer to a tow vehicle, said trailer hitch system comprising: a shank attachable to said trailer, said shank having a nose at a terminal end of the shank and a mid-shank extending in a substantially horizontal direction immediately adjacent to the nose, said nose and mid-shank being configured to be received in a receiver attachable to said tow vehicle and a vertical adjustment extending in a substantially vertical direction; and a depth stop connected to the shank and configured to abut said receiver to limit a depth said shank extends within said receiver; wherein said mid-shank includes a step reduction having a corner which decreases a cross-sectional area of the mid-shank immediately adjacent to the nose and an alignment indicator provided on an external surface and extending along the external surface of the mid-shank in a substantially horizontal direction; wherein alignment indicator is a color that is different than a color of the mid-shank; and wherein the nose is configured to reduce in cross-sectional area toward the terminal end of the shank.
2. The trailer hitch system of claim 1, wherein the alignment indicator is substantially linear.
3. The trailer hitch system of claim 1, wherein the alignment indicator extends along a top surface of the mid-shank.
4. The trailer hitch system of claim 1, wherein the alignment indicator extends along a side surface of the mid-shank.
5. The trailer hitch system of claim 1, wherein the alignment indicator is recessed into the mid-shank.
6. The trailer hitch system of claim 1, wherein the alignment indicator is more reflective than the mid-shank.
7. The trailer hitch system of claim 1, wherein said depth stop is a collar that is adjustably coupled to the mid-shank enabling the depth stop to change position with respect to the mid-shank.
8. The trailer hitch system of claim 1, wherein the corner of the step reduction is a right-angled corner.
9. A trailer hitch system for attaching a trailer to a tow vehicle, said trailer hitch system comprising: a shank attachable to said trailer, said shank having a nose at a terminal end of the shank and a mid-shank extending in a substantially horizontal direction immediately adjacent to the nose, said nose and mid-shank being configured to be received in a receiver attachable to said tow vehicle, and a vertical adjustment extending in a substantially vertical direction; a depth stop adjustably connected to the mid-shank and configured to abut said receiver, to limit a depth said shank extends within said receiver, wherein said depth stop is a collar that is adjustably coupled to the mid-shank enabling the depth stop to change position with respect to the mid-shank; and wherein said mid-shank includes a step reduction having a corner which decreases a cross-sectional area of the mid-shank immediately adjacent to the nose and wherein the mid-shank includes at least two alignment indicators provided on at least two respective external surfaces of the mid-shank, each of the at least two alignment indicators extending along the mid-shank in a substantially horizontal direction, and wherein at least one of the alignment indicators is a color that is different than a color of the mid-shank.
10. The trailer hitch system of claim 9, wherein each of the alignment indicators are substantially linear.
11. The trailer hitch system of claim 9, wherein at least one of the alignment indicators extends along a top external surface of the mid-shank.
12. The trailer hitch system of claim 9, wherein at least one of the alignment indicators extends along a side external surface of the mid-shank.
13. The trailer hitch system of claim 9, wherein at least one of the alignment indicators is recessed into the external surface of the mid-shank.
14. The trailer hitch system of claim 9, wherein at least one of the alignment indicators is more reflective than the mid-shank.

15. The trailer hitch system of claim 9, wherein the corner of the step reduction is a right-angled corner.

16. A trailer hitch system for attaching a trailer to a tow vehicle, said trailer hitch system comprising: a shank attachable to said trailer, said shank having a nose at a terminal end of the shank and a mid-shank extending in a substantially horizontal direction immediately adjacent to the nose, said nose and mid-shank being configured to be received in a receiver attachable to said tow vehicle; a depth stop connected to the shank and configured to abut said receiver to limit a depth said shank extends within said receiver; wherein said mid-shank includes a step reduction having a corner which decreases a cross-sectional area of the mid-shank immediately adjacent to the nose and an alignment indicator on an external surface of the shank configured to provide a visual indicator of the orientation of the shank to a back-up camera of the tow vehicle; wherein the nose is configured to reduce in cross-sectional area toward the terminal end of the shank, and wherein the alignment indicator is a color that is different than a color of the nose.

17. The trailer hitch system of claim 16, wherein the alignment indicator extends along a side surface of the mid-shank.

18. The trailer hitch system of claim 16, wherein the alignment indicator is more reflective than the nose.

19. The trailer hitch system of claim 16, wherein the alignment indicator is substantially linear.

20. The trailer hitch system of claim 16, wherein the corner of the step reduction is a right-angled corner.

21. A trailer hitch system for attaching a trailer to a tow vehicle, said trailer hitch system comprising: a shank attachable to said trailer, said shank having a nose at a terminal end of the shank and a mid-shank having a length and extending in a substantially horizontal direction immediately adjacent to the nose, said nose and mid-shank being configured to be received in a receiver attachable to said tow vehicle, wherein the mid-shank includes a top surface, a bottom surface, and side surfaces; and at least two alignment indicators, including a first alignment indicator extending along at least a portion of a top external-facing surface and at least a second alignment indicator extending along at least a portion of a side external surface of the mid-shank, each of the at least two alignment indicators extending in a substantially horizontal direction to provide a visual indicator of the orientation of the shank to a back-up camera of the tow vehicle; wherein the at least two alignment indicators are a color that is different than a color of the nose.

22. The trailer hitch system of claim 21, wherein the at least two alignment indicators are substantially linear.

23. The trailer hitch system of claim 21, wherein the at least two alignment indicators extends along each side external surface of the mid-shank.

24. The trailer hitch system of claim 21, wherein the at least two alignment indicators are more reflective than the mid-shank.

25. The trailer hitch system of claim 21, wherein said mid-shank includes a step reduction which decreases a cross-sectional area of the mid-shank immediately adjacent to the nose.

26. The trailer hitch system of claim 21, further comprising: a depth stop connected to the shank and configured to abut said receiver to limit a depth said shank extends within said receiver.
